MARKET TWIN

Why Simulation Results Can Vary Between Runs

Understanding result variance
and using it correctly in your decisions

Market Twin — Result Interpretation Guide
Mr.AI Inc. · 2026.06

! At a glance

> **Market Twin is a probabilistic simulation, not a deterministic calculator.** Running the same product with the same inputs again can produce slightly different scores and rankings — and occasionally a different recommended country altogether. This is **not a bug; it is the nature of simulation**. The report never hides this variation — it surfaces it through a **variance grade, Top-2 tied candidates, and a consensus rate**. So results should be read as a **ranking, range, and confidence level — not as a single fixed "answer."**

1 Why does the same input give different results?

There are five causes, and most of them have **nothing to do with language**.

① The LLM's probabilistic generation (the biggest factor)

The simulation uses a large language model (LLM) to reason about hundreds of virtual consumers. An LLM generates answers probabilistically (temperature), so it produces a **slightly different response every time**, even to the same prompt. These small differences accumulate from persona reactions up into country scores.

② Persona sampling

Each run **re-samples** the personas according to the profession / income / age distribution. Which personas are drawn shifts how much weight favorable vs. critical voices carry for a given market.

③ When candidate countries are closely matched

If the true attractiveness of the candidate countries is **nearly equal**, even a tiny fluctuation can flip the #1 and #2. Conversely, when one country dominates, you get the same result every time. **Result instability is itself information — it tells you the candidates are neck-and-neck.**

④ The strength of the input signal

If the product description doesn't carry enough cues pointing to a specific market (target, channel, cultural codes), the model can't confidently distinguish markets and results swing more. **The more specific the input, the more stable the result.**

⑤ (Secondary) the output language

The output language (Korean, English, etc.) is only a **weak variable** that can slightly change wording and nuance. Language itself does not determine the result — **factors ①–④ have a far larger effect.**

2 This is "by design," not a flaw

Real-world market response is inherently uncertain. Declaring a single "answer" from one run would actually be dangerous. That is why Market Twin deliberately:

- runs **multiple simulations (an ensemble)** and aggregates them,
- **measures and reports the variation** (variance) between them, and
- does **not force a single winner** when the race is close (Top-2 tied candidates).

> Key idea: variance is **not a flaw to hide — it is information you need**. It tells you "how much you can trust this call."

3 Four ways the report surfaces variation

Device	Meaning / how to read it
Variance grade	LOW stable · MODERATE normal · HIGH very volatile. HIGH signals "don't trust a single result as-is."
Top-2 tied candidates	When the #1 / #2 score gap is small, the report presents **both countries** instead of forcing one winner. It means "evaluate both."
Consensus %	The share of sims / models that **agreed on the same conclusion**. Higher = more reliable. (e.g. 67% = 2 of 3 reached the same call.)
Confidence	**STRONG / MODERATE / WEAK**. WEAK indicates a dispersed result — consider further validation or stronger inputs.

4 Interpretation guide by variance level

Symptom	Interpretation	What to do
Score moves a few points, **ranking holds**	Normal variation	Trust it — safe to ignore
#1 / #2 **occasionally swap** / shown as Top-2	The two markets are effectively tied	Keep **both** and pilot them in parallel
Variance HIGH / a **different country** wins each run	Markets are hard to distinguish (neck-and-neck or weak input signal)	Narrow the candidates · strengthen inputs · rerun at a higher tier

5 A real example — same input, different #1

Here is an actual result from validating a vape (pod-system) product at the Hypothesis tier (3 sims). **With identical inputs, the three sims each picked a different country as #1.**

sim	Country picked as #1
sim 1	Japan (JP)
sim 2	United Kingdom (GB)
sim 3	Indonesia (ID)

The vote split **JP 33% · GB 33% · ID 33%** — completely even. Variance was **HIGH** and confidence was **WEAK**. At that point the report **warned on its own**:

> "The same fixture produces very different country scores per run. Trust the ensemble — a single sim alone would be unreliable."

For this product, the nicotine regulation / Gen-Z / collectible traits play out **similarly across many countries**, so the model could not confidently pick a #1. **This was not caused by language — the input itself failed to distinguish markets.** Such a result should be read not as "this one country," but as **"several candidates are close — narrow them down and look again."**

6 How to improve reliability

1. **Narrow the candidate countries.** Comparing too many at once makes them hard to distinguish. Compressing to 3–5 stabilizes the ranking.
2. **Make the input specific.** Spell out target demographic, channels, and regional/cultural codes to strengthen the signal. (But don't over-assert one market, or you introduce bias.)
3. **Run at a higher tier.** More sims means a more stable aggregate and lower variance.
4. **Always read variance and consensus together.** Don't look at the score alone — check "how trustworthy is this."
5. **If Top-2 appears, evaluate both.** Piloting two markets in parallel is safer than forcing one choice.

7 Decision checklist

- ☐ Did you check the **variance grade**, not just the recommended country?
- ☐ Did you read **consensus % and confidence** (STRONG/WEAK) together?
- ☐ If a **Top-2 tie** is shown, did you evaluate both markets?
- ☐ If variance is HIGH, did you **rerun** after narrowing candidates or strengthening inputs?
- ☐ Did you treat the result as a **hypothesis / priority**, not a verdict?

> **In short:** the value of Market Twin is not "picking the answer for you" — it is **seeing the priority and uncertainty of your market hypotheses, with data, before you launch.** If results wobble, that wobble itself is an important signal to "tread carefully here."